

MERCHANT API DOCUMENT

WITHDRAW/FUND OUT

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Document Release Note

Version	Date	Description	Author
0.1.0	16 May 2020	First Draft	K5
1.0.0	30 Oct 2020	Update the changes according to the latest upgrade in API	K5
1.0.1	27 Jan 2021	Add Java version of Message Encryption Example	K5
1.0.2	29 Jan 2021	Updated Testing API endpoint	K5

Prerequisite

Merchant Account

When merchant account is ready, we will provide merchant the followings

1) Merchant Back Office URL
2) APIEndpoint
3) Merchant ID (16 characters, alphanumeric unique ID to identify merchant)
4) Merchant Key (16 characters, secret key for encryption)
5) Merchant Hash IV (16 characters, the hash IV (Initiation Vector) used for encryption)

Fund Out (Withdrawal) Request

Merchant Back Office : http://merchant-test.primepayco.com/
UATAPI Endpoint : http://api-test.primepayco.com/api/merchant/transact
HTTP Method : POST
HTTP Header : Content-Type: application/json
Request Data Format : JSON
Response Data Format : JSON

Request Parameters

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	This will be provided	
message			Refer Message details in the table below. The message in json format is encrypted json data, refer the **Appendix A - Message** Encryption section below for the data fields	

Message (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
merchantId		Char (16)	This will be provided	
transactionId		Char (45)	A unique reference from merchant to identify the fund out/withdrawal request transaction.	
transactionType		Char (1)		Possible Value: **W**: Withdraw
currency		Char (3)	Currency Code (ISO 4217)	MYR

Field ID	Field Name	Data Type	Description	Example Value
amount		Decimal(20, 2)		
callback		URL	When request is completed, a confirmation request will be sending to the callback URL	
bankName		Char (45)	Designated Bank Details for the requested amount to transfer.	
bankAcctName		Char (45)		
bankAccNum		Char (45)		
bankBranch		Char (45)		
Optional Fields				
transactionDesc		Char(100)	Description for the transaction	
userId		Char (45)	The user/player who owns the designated bank details	
userName		Char (45)		
userPhone		Char (20)		
userEmail		Char (255)		
userIp		Char (20)		

Example

Body Data (JSON Format)


```
{
  "merchantId": "82164414b30aa575",
  "message":
"N+uVCc2MrpWk+rr2kaQ1pXyZqr4uS4v0Rhk7TsfezgRKNlqgJDQV6qpxxaJuY2xRUb7ovvwcIXedON7vCDkXsh11Zd1iSjdlA0TXxmw011eZycDSEyaWYm
af4E4YciNwx6lDs1Nb/l2uZyqTYqZJxp4T9VgMuDHlKsQxrGOWNLXgiE2243K2ZTriYr95iPLLAchUSXDzekpWOLS/XBHQVtnx1q/wWLIPEfmmgUhxoG0i1
EzS7W4gjOoIWjuI0OqKF4MCyMJcd+FjuQW6q2KSdC7GR/cXtjt3vpZ3b1DFnj8dvJFZ/4jKqFy8KzgpsBCBYJg+Fqx/BmtiUHn5H6/iNMRICxc8RhCxB2E1
sGQcIKYxQ6vA7sr9USm8S66gkogd"
}
```

Message before encrypted


```
{
  "merchantId":"82164414b30aa575",
  "transactionId":"20200318011925903",
  "transactionType":"W",
  "currency":"MYR",
  "amount":"1000",
  "callback":"http://payment-gateway/callback",
  "bankName":"Test Bank",
  "bankAcctName":"Test A",
  "bankAccNum":"111111111",
  "bankBranch": "KL"
}
```

Response Parameters

Field ID	Field Name	Data Type	Description	Example Value
status		Char (10)		success or fail
referenceId		Char (45)	The unique fund out/withdrawal request transaction ID from us for your reference	When status = success
errors		Text		When status = fail

Example

Response Data when the request is success

```
{
  "status":"success”,
  "referenceId": "e4f3c568-8ec1-48bf-bd39-b299467d76b0"
}
```

Response Data when request is failed

```
{
  "status": "fail",
  "errors": "error message"
}
```

Callback URL Request

We will inform merchant about the status change of the fund out / withdrawal request via the Callback URL. We may send the request multiple times and Merchant is required to handle the following situations, when fund out / withdrawal request is

1) completed/success–the requested fund out amount is successfully transferred to the designated bank account, but there could be the following situations:
    a. refunded –For non instant transfer, requested fund may be transferred successfully initially, but a **bank reversal** could takeplace due to some reasons.
    b. completed with adjustment –service fee / transaction fee could be adjusted by us due to error input.

2) rejected – due to incorrect designated bank details or insufficient fund balance

HTTP Method : POST
HTTP Header : Content-Type: application/json
Request Data Format : JSON
Response Data Format : JSON

Request Parameters

Field ID	Field Name	Data Type	Description	Example Value
merchantId		Char (16)	The unique merchant account id	
transactionId		Char (45)	The unique reference from merchant to identify the fund out/withdrawal request transaction.	
message		Text	Refer Message details in the table below. The message in json format is encrypted json data, refer the **JAVA** 8 **import javax.crypto**.Cipher; import javax.crypto.SecretKey; import javax.crypto.spec.IvParameterSpec; import javax.crypto.spec.SecretKeySpec; import javax.crypto.IllegalBlockSizeException;	

Field ID	Field Name	Data Type	Description	Example Value

			import javax.crypto.BadPaddingException;
			import javax.crypto.NoSuchPaddingException;
			import java.security.NoSuchAlgorithmException;
			import java.security.InvalidKeyException;
			import
			java.security.InvalidAlgorithmParameterException;
			import java.io.UnsupportedEncodingException;
			import java.util.Base64;
			
			String content = "{" +
			"\"merchantId\": \"94b2522ba609909d\"," +
			"\"transactionId\": \"20201030023201612\"," +
			"\"transactionType\": \"D\"," +
			"\"payMethod\": \"offline\"," +
			"\"currency\": \"RMB\"," +
			"\"amount\": \"500.00\"," +
			"\"callback\":
			\"https://enmu1r0iut2ws.x.pipedream.net\"," +
			"\"response\":
			\"https://enmu1r0iut2ws.x.pipedream.net\"" +
			"}";
			
			String secretKey = "abcdefghhgfedcba";
			String secretIv = "1234567887654321"; //Merchant
			Pass1 + Pass2
			
			try {
			Cipher cipher =
			Cipher.getInstance("AES/CBC/PKCS5Padding");
			SecretKeySpec keySpec = new
			SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
			IvParameterSpec ivSpec = new
			IvParameterSpec(secretIv.getBytes("utf-8"));
			cipher.init(Cipher.ENCRYPT_MODE, keySpec,
			ivSpec);
			byte[] ciphertext =
			cipher.doFinal(content.getBytes("UTF-8"));
			String message =
			Base64.getUrlEncoder().encodeToString(ciphertext);	

Field ID	Field Name	Data Type	Description	Example Value

			<pre> System.out.println("message=" + message);
			} catch (NoSuchAlgorithmException e) {
			  System.err.println(e.getMessage());
			} catch (NoSuchPaddingException e) {
			  System.err.println(e.getMessage());
			} catch (InvalidAlgorithmParameterException e) {
			  System.err.println(e.getMessage());
			} catch (BadPaddingException e) {
			  System.err.println(e.getMessage());
			} catch (UnsupportedEncodingException e) {
			  System.err.println(e.getMessage());
			} catch (InvalidKeyException e) {
			  System.err.println(e.getMessage());
			} catch (IllegalBlockSizeException e) {
			  System.err.println(e.getMessage());
			}</pre>
			Appendix B - Message Decryption section below to descript the string in order to obtain the json format data.	

Message (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
referenceId		Char (45)	The unique fund out/withdrawal request transaction ID from us for merchant reference	
status		Integer	The request status	Possible values **1**: Completed **2**: Rejected **4**: Refunded
statusDesc		Char (20)	The description of the request status	When status value is

Field ID	Field Name	Data Type	Description	Example Value
				1 -**Completed** 2 -**Rejected** 4 - **Refunded**
requestDate		DATETIME	The fund out / withdrawal request date Date format: YYYY-MM-DD HH:MM:SS	2020-05-16 18:01:08
transactionType		Char (1)	The transaction type	Possible Value: **W**: Withdraw
transactionTypeDesc		Char (20)	The description of the transaction type	When transaction type is W - **Withdraw**
currency		Char (3)	Currency Code (ISO 4217). As per the currency from the request	MYR
amount		Decimal (20, 2)	As per the amount from the request	1000.00
transactionDate		DATETIME	The date of status changed Date format: YYYY-MM-DD HH:MM:SS	2020-05-16 18:01:08
remark		Char (255)	The remark input by us	
When status = "1" (Completed)				
frombankName		Char (45)	The source bank name transferring the requested amount	
bankTransactionId		Char (45)	The bank reference id of the fund transfer	
transactionAmount		Decimal (20, 2)	The actual amount of the fund transferred	
transactionFee		Decimal (20, 2)	The	
serviceFee		Decimal (20, 2)		
adjustment			When it is completed but there is input error of service fee/transaction fee.	

Field ID	Field Name	Data Type	Description	Example Value
			Refer to the Adjustment section below	
When status = "4" (Refunded)				
refundedReason		Char (255)	The reason of the reverse of the fund out.	

Adjustment (in JSON Format)

Field ID	Field Name	Data Type	Description	Example Value
Mandatory				
adjustedDate		DATETIME	The date that we make the adjustment Date format: YYYY-MM-DD HH:MM:SS	
adjustedReason		Char (255)	The reason of the adjustment	
Conditional				
serviceFeeBeforeAdjusted		Decimal (20, 2)	This will return the previous value of the service fee when the adjustment is on service fee. The new value is replaced in the **serviceFee** above.	
transactionFeeBeforeAdjusted		Decimal (20, 2)	This will return the previous value of the transaction fee when the adjustment is on transaction fee. The new value is replaced in the **transactionFee** above.	

Example

Request Post Data

```
{
  "merchantId":"82164414b30aa575",
  "transactionId":"20200504054453805",
```

```
  "message":
"Eis\/kISbtVtkLckK\/LHMQQ1kZ9yy0oZV3tk0CPDJv5xfT+xPvbDT510cPL4b8QedGBsWkHqRmpDfbIGyWZVSB3egVdEEXW6I8q6vxhccIoHruCclCzrI
vP9e\/Y1ick1gkMAEZRCTs5jLLvD3waDU+nzMGIyY2oGtn0V0Lu+bSVAjigRDBrwxnullPU2ABvzh9jq4ARAwxjJQPBVIAAeaWTNsqRi1dvMo4xp6Oq1i9r
Uafg4DREkizh\/h4p+1a+YwvluuiMnZC4RcCHsiW5CyjPd+Tg6bSkEYFm4Ddh8bNgLgxxlM9U7BTwX2mvSRnLoZotTgflVAva20LiGcPeM3AllcPuma5khd
lAmxaTnFc3uaoKtWPAPEElgldcEZvh\/2Mlra+2t2MfegpcDpquFwr0+Fm+9ZUg2zQUjAx7ry3N6DuFMlKMTJqxs3\/d\/EXvqI\/gL8K9ntXMnC229140u
71UvZyv+jfDr7mgsSwurW5rsIDpZadwKezEsv7Y7KBe9xGPuh+jNev+SqJKytvoG6xwttnqy+K6IPiLFLwXs54GlhKizDVnIamp2cJpKeGeH+V7C84jzX5f
+dXf+yTDkAffUjpD4b0heJM\/7PMwdZqbFho7KHKZyR9OyHkXcHa2J2jN8FB\/egBh0PWncQutxswGUk1+maQWdZkZM5KzF0SRc50kCvGZPi2vgvKZMcybn
XfOttO2Z+PTmd5b0nOtuo5w=="}
```


Decrypted Message:


```
{
  "referenceId": "77b90267-1c71-4274-9316-0d3b4c4313fb",
  "status": 1,
  "statusDesc": "Completed",
  "requestDate": "2020-05-04 05:44:55",
  "transactionType": "W",
  "transactionTypeDesc": "Withdraw",
  "currency": "RMB",
  "amount": "500.00",
  "transactionDate": "2020-05-04 08:53:12",
  "remark": "",
  "frombankName": "上海浦东发展银行",
  "bankTransactionId": "324324323",
  "transactionAmount": "500.00",
  "transactionFee": "2.00",
  "serviceFee": "1.00",
  "adjustment": {
    "adjustedDate": "2020-05-04 07:51:52",
    "adjustedReason": "amended service fee",
    "serviceFeeBeforeAdjusted": "0.00"
  }
}
```


Response Parameters

Field ID	Field Name	Data Type	Description	Example Value
status		Char (10)		**success** or **fail**

Field ID	Field Name	Data Type	Description	Example Value
message		Char (255)	Description for Merchant reference	

Example

Response JSON

```
{
  "status":"success",
  "message":""
}
```

Appendix A - Message Encryption

Encryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Encrypt the message string (JSON Format) with AES128, cipher mode CBC
2) Use Base64 to encode the AES128 encrypted text

Example

PHP 7

```
$content = '{
      "merchantId":"82164414b30aa575",
      "transactionId":"20200318011925903",
      "transactionType":"W",
      "currency":"MYR",
      "amount":"1000",
      "callback":"http://payment-gateway/callback",
      "bankName":"Test Bank",
      "bankAcctName":"Test A",
      "bankAccNum":"111111111",
      "bankBranch":"KL"
}';

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = openssl_encrypt($content, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
$message = base64_encode($encrypted);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"transactionId\": \"20201030023201612\"," +
  "\"transactionType\": \"D\"," +
  "\"payMethod\": \"offline\"," +
  "\"currency\": \"RMB\"," +
  "\"amount\": \"500.00\"," +
  "\"callback\": \"https://enmu1r0iut2ws.x.pipedream.net\"," +
  "\"response\": \"https://enmu1r0iut2ws.x.pipedream.net\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.ENCRYPT_MODE, keySpec, ivSpec);
  byte[] ciphertext = cipher.doFinal(content.getBytes("UTF-8"));
  String message = Base64.getUrlEncoder().encodeToString(ciphertext);

  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
```

```
} catch (NoSuchPaddingException e) {
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```

Appendix B - Message Decryption

Decryption Method : AES128
Block Cipher Mode : CBC

Merchant will be provided the followings when merchant account is ready

1) Merchant Key (16 characters)
2) Merchant Hash IV (16 characters)

1) Use Base64 to decode the message to return raw message in bytes
2) Decrypt the raw message in bytes with AES128, cipher mode CBC to obtain the JSON plain data.

Example

PHP 7

```
$request = '{
      "merchantId":"82164414b30aa575",
      "transactionId":"20200504054453805",
      "message":"Eis\/kISbtVtkLckK\/LHMQQ1kZ9yy0oZV3tk0CPDJv5xfT+xPvbDT510cPL4b8QedGBsWkHqRmpDfbIGyWZVSB3egVdEEXW6I8q6v
xhccIoHruCclCzrIvP9e\/Y1ick1gkMAEZRCTs5jLLvD3waDU+nzMGIyY2oGtn0V0Lu+bSVAjigRDBrwxnullPU2ABvzh9jq4ARAwxjJQPBVIAAeaWTNsqR
i1dvMo4xp6Oq1i9rUafg4DREkizh\/h4p+1a+YwvluuiMnZC4RcCHsiW5CyjPd+Tg6bSkEYFm4Ddh8bNgLgxxlM9U7BTwX2mvSRnLoZotTgflVAva20LiGc
PeM3AllcPuma5khdlAmxaTnFc3uaoKtWPAPEElgldcEZvh\/2Mlra+2t2MfegpcDpquFwr0+Fm+9ZUg2zQUjAx7ry3N6DuFMlKMTJqxs3\/d\/EXvqI\/gL
8K9ntXMnC229140u71UvZyv+jfDr7mgsSwurW5rsIDpZadwKezEsv7Y7KBe9xGPuh+jNev+SqJKytvoG6xwttnqy+K6IPiLFLwXs54GlhKizDVnIamp2cJp
KeGeH+V7C84jzX5f+dXf+yTDkAffUjpD4b0heJM\/7PMwdZqbFho7KHKZyR9OyHkXcHa2J2jN8FB\/egBh0PWncQutxswGUk1+maQWdZkZM5KzF0SRc50kC
vGZPi2vgvKZMcybnXfOttO2Z+PTmd5b0nOtuo5w=="
}';

$content = json_encode($request);
$content = $content->message;

$secret_key = $merchantKey
$secret_iv = $merchantHashIV;
$mode = "AES-128-CBC";
$encrypted = base64_decode($content);
$message = openssl_decrypt($encrypted, $mode, $secret_key, OPENSSL_RAW_DATA, $secret_iv);
```

JAVA 8

```
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import javax.crypto.IllegalBlockSizeException;
import javax.crypto.BadPaddingException;
import javax.crypto.NoSuchPaddingException;
import java.security.NoSuchAlgorithmException;
import java.security.InvalidKeyException;
import java.security.InvalidAlgorithmParameterException;
import java.io.UnsupportedEncodingException;
import java.util.Base64;

String requestMessage =
  "dO_ih-LJC7fpBjKRTr0Hbmw0rPvn4MKP3XwDPWxPYFY7T09dEze-tar7NXdYqJ0kY-osQs822lyks-
SI3s9NCidEtbulRDtDeBF2Tq0lhMAGseEDgJnyHu6CtSTLCM3iecKYX9qfsK64GndX1wrIAL2mMPeiWNfYN14rowLliqQU8TkvVR2Rd6Ml2XpecL5YXAUsG
Tq8omxFb3YBB5aYoU9n1rh-NzL_43IQJQw-Tx4Kc-eLniTwNJgVUbpYox3vFjrbVeF3pNFZ832nWnXB7RZ0FLCdz77CUXJ4sQdS17z7JgwXmCPxD22-
YybQixlAbP-tSPIebs1i44qiSo7b9asG7d-y20A7ePk3qgLj_PQ=";

String content = "{" +
  "\"merchantId\": \"94b2522ba609909d\"," +
  "\"message\": \"" + requestMessage + "\"" +
"}";

String secretKey = "abcdefghhgfedcba";
String secretIv = "1234567887654321"; //Merchant Pass1 + Pass2

try {
  Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
  SecretKeySpec keySpec = new SecretKeySpec(secretKey.getBytes("utf-8"), "AES");
  IvParameterSpec ivSpec = new IvParameterSpec(secretIv.getBytes("utf-8"));
  cipher.init(Cipher.DECRYPT_MODE, keySpec, ivSpec);
  byte[] requestMessageBytes = Base64.getUrlDecoder().decode(requestMessage);
  byte[] plainText = cipher.doFinal(requestMessageBytes);
  String message = new String(plainText );

  System.out.println("message=" + message);
} catch (NoSuchAlgorithmException e) {
  System.err.println(e.getMessage());
```

```
} catch (NoSuchPaddingException e) {
  System.err.println(e.getMessage());
} catch (InvalidAlgorithmParameterException e) {
  System.err.println(e.getMessage());
} catch (BadPaddingException e) {
  System.err.println(e.getMessage());
} catch (UnsupportedEncodingException e) {
  System.err.println(e.getMessage());
} catch (InvalidKeyException e) {
  System.err.println(e.getMessage());
} catch (IllegalBlockSizeException e) {
  System.err.println(e.getMessage());
}
```